

Detailed Software Requirement Specification for Community Clinic Appointment and Inventory System of Rural Siddha Hospital Thellipalai

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Contents

1. Introduction	4
1.1 Purpose.....	4
1.2 Summary.....	4
1.3 Company Overview.....	4
1.4 Project Overview.....	4
1.5 Scope.....	4
1.6 Assumptions	5
1.7 Definitions, Acronyms and Terminology	5
<i>REFERENCES</i>	5
2. Project Scope and Impact.....	7
2.1 Scope Inclusions	7
2.2 Scope Exclusions.....	7
3. Functional Requirements (FRs)	8
3.1 Core Functionalities.....	8
3.1.1 User Registration.....	8
3.1.2 User Login & Authentication.....	8
3.1.3 Book Appointment.....	9
3.1.4 Cancel Appointment.....	9
3.1.5 View Appointment Schedule & Patient Records	10
3.1.6 Record Prescription	10
3.1.7 Dispense Medicine	11
3.1.8 Manage Inventory	11
3.1.9 Send Order Request	12
3.1.10 Generate Reports	12
3.1.11 System Administration.....	12
3.2 Data Archival and Retention	13
3.3 User Profiles, Roles, and Privileges	13
3.4 Reporting Requirements	13
4. Non-Functional Requirements.....	14
4.1.1 Performance and Load Requirements	14
4.1.2 Compatibility Requirements	14

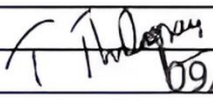
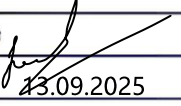
4.1.3	External Interface Requirements	15
4.1.4	Security and Authentication requirements	28
4.1.5	Quality Assurance Requirements	29
4.1.6	Development Requirements	30
4.1.7	Deployment Requirements	32
4.1.8	Special Documentation Requirements	33
4.1.9	Applicable Standards.....	33
4.1.10	On-line User Documentation and Help System Requirements	33
4.1.11	Usability Requirements	34
Appendix		34

Document Revisions

Date	Version	Description	Author

Document Approval

Quality Software Corporation and Dr.T.Thileepan have reviewed this document and hereby agree that the contents herein are accurate. Any changes to this document must be communicated in writing and signed off by both parties.

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1. Introduction

1.1 Purpose

This document specifies the requirements for the **Community Clinic Appointment and Inventory System**. The purpose of this system is to provide a web-based solution that streamlines appointment scheduling, manages patient and doctor information, and monitors inventory for medical supplies at Rural Siddha Hospital, Tellipalai. This SRS will serve as a foundation for the design, development, and testing phases of the project.

1.2 Summary

Currently, appointments and inventory are managed manually at the hospital, causing inefficiencies such as scheduling conflicts, long waiting times, and stock shortages. The proposed system aims to eliminate these issues by providing an online, centralized, and user-friendly platform.

1.3 Company Overview

Rural Siddha Hospital, Tellipalai is a community-focused healthcare institution in the Northern Province of Sri Lanka. The hospital provides traditional Siddha medicine alongside outpatient treatments for local residents.

1.4 Project Overview

The project involves the design and development of a **web-based application**. It will support patients, doctors, staff, and administrators with features like online appointment booking, schedule management, inventory control, and reporting.

1.5 Scope

This document covers all functional and non-functional requirements for the system, including assumptions and limitations. It defines what the system will include and what is excluded from its scope.

1.6 Assumptions

- Users will have access to the internet and basic computing devices (PCs or smartphones).
- The hospital will provide accurate initial data (doctor schedules, inventory levels).
- Patient data will be stored in compliance with local data protection policies.

1.7 Definitions, Acronyms and Terminology

- **SRS** – Software Requirements Specification
- **UI** – User Interface
- **RBAC** – Role-Based Access Control

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2. Project Scope and Impact

2.1 Scope Inclusions

- Online appointment scheduling and cancellation.
- Real-time doctor schedule management.
- Inventory tracking with low-stock alerts.
- Centralized administration for users and reports.

2.2 Scope Exclusions

- No mobile app in the first release (web application only).
- No integration with third-party hospital ERP systems.

3. Functional Requirements (FRs)

This section defines the **functional requirements** of the Community Clinic Appointment and Inventory Management System. Each function is explained in detail, covering purpose, process flow, user roles, input/output, and error handling.

3.1 Core Functionalities

3.1.1 User Registration

- **Actors:** Patient, Receptionist
- **Description:** Allows new patients to register in the system. Receptionists may also register patients who cannot self-register.
- **Process Flow:**
 1. User enters personal details (Name, Age, Gender, Contact Info, Address).
 2. System validates input (mandatory fields, unique email/phone).
 3. A new account is created, stored in the database.
 4. Confirmation displayed.
- **Error Handling:** Duplicate email/phone, invalid data format, missing mandatory details.
- **Output:** Successful registration → patient profile created.

3.1.2 User Login & Authentication

- **Actors:** Patient, Receptionist, Doctor, Pharmacist, Administrator
- **Description:** Provides secure login and authentication.
- **Process Flow:**
 1. User enters username and password.
 2. System validates credentials.
 3. If valid → user redirected to dashboard based on role.
- **Error Handling:** Invalid credentials, multiple failed attempts (account lockout).
- **Output:** Secure user session with appropriate privileges.

3.1.3 Book Appointment

- **Actors:** Patient, Receptionist
- **Includes:** Check Doctor Availability, Send Notifications
- **Extends:** Process Payment (if required)
- **Description:** Enables booking of doctor appointments.
- **Process Flow:**
 1. User selects doctor and preferred time slot.
 2. System checks doctor availability.
 3. If slot available → appointment booked.
 4. If required, payment is processed.
 5. Notifications sent to patient and doctor.
- **Error Handling:** Slot unavailable, payment failure, system error.
- **Output:** Appointment confirmed and logged in system.

3.1.4 Cancel Appointment

- **Actors:** Patient, Receptionist
- **Includes:** Send Notifications
- **Description:** Allows users to cancel scheduled appointments.
- **Process Flow:**
 1. User selects appointment from schedule.
 2. System cancels appointment and updates database.
 3. Notifications sent to doctor and patient.
- **Error Handling:** Invalid appointment ID, already canceled appointment.
- **Output:** Appointment removed from schedule.

3.1.5 View Appointment Schedule & Patient Records

- **Actors:** Patient, Receptionist, Doctor
- **Description:** Enables viewing of appointment schedules and patient medical history.
- **Process Flow:**
 1. User requests appointment schedule or patient record.
 2. System retrieves and displays relevant data.
- **Error Handling:** Unauthorized access, missing record.
- **Output:** Schedule or patient details displayed.

3.1.6 Record Prescription

- **Actors:** Doctor
- **Description:** Doctors record prescriptions for patients.
- **Process Flow:**
 1. Doctor opens patient record.
 2. Doctor enters prescription details (medicine name, dosage, instructions).
 3. System stores prescription in database.
- **Error Handling:** Invalid drug name, incomplete prescription.
- **Output:** Prescription stored and linked to patient record.

3.1.7 Dispense Medicine

- **Actors:** Pharmacist
- **Includes:** Prescription Validation
- **Description:** Pharmacists dispense medicines as per doctor's prescription.
- **Process Flow:**
 1. Pharmacist retrieves patient prescription.
 2. Medicine dispensed if available in stock.
 3. System updates inventory.
- **Error Handling:** Out-of-stock medicine, mismatch with prescription.
- **Output:** Medicine dispensed, stock updated.

3.1.8 Manage Inventory

- **Actors:** Pharmacist
- **Extends:** Stock Low
- **Includes:** Generate Reports
- **Description:** Manages medicine stock (add, update, delete).
- **Process Flow:**
 1. Pharmacist updates inventory after each dispense/order.
 2. System checks stock levels.
 3. If stock is low, an alert is generated.
- **Error Handling:** Incorrect stock entry, duplicate medicine entry.
- **Output:** Updated and accurate inventory records.

3.1.9 Send Order Request

- **Actors:** Pharmacist
- **Description:** Send orders medicines to administrator when stock is low.
- **Process Flow:**
 1. Pharmacist identifies low-stock medicine.
 2. Creates and sends order request.
 3. Order logged in the system.
- **Error Handling:** Duplicate orders, invalid medicine ID.
- **Output:** Supplier order confirmation.

3.1.10 Generate Reports

- **Actors:** Administrator, Pharmacist
- **Description:** Provides detailed reports on appointments, prescriptions, and inventory.
- **Types of Reports:**
 - Appointment reports (daily/weekly/monthly trends).
 - Patient medical reports.
 - Inventory status reports.
 - System audit reports.
- **Error Handling:** Missing data, report formatting errors.
- **Output:** Reports in PDF/printable format.

3.1.11 System Administration

- **Actors:** Administrator
- **Description:** Admin manages users, doctors, and system operations.
- **Functions Include:**
 - **Maintain User Accounts:** Add, delete, or update user accounts.
 - **Update Doctors:** Add/delete doctors, update specialties and experience.
 - **View User Information:** Retrieve user details.

- **Backup & Restore Data:** Ensure data safety and recovery.
- **Audit Logs:** Track system activities.
- **Send Reminders:** Notify patients of upcoming appointments.
- **Error Handling:** Backup failure, unauthorized access.
- **Output:** Secure and smooth system operation.

3.2 Data Archival and Retention

- Patient records retained for **minimum 5 years**.
- Appointment logs retained for **2 years**.
- Archived data must be encrypted and retrievable securely.

3.3 User Profiles, Roles, and Privileges

- **Patient:** Register, login, book/cancel appointment, view schedule & records.
- **Receptionist:** Book/cancel appointment on behalf of patients, view schedules.
- **Doctor:** View schedules, patient records, record prescriptions.
- **Pharmacist:** Dispense medicines, manage inventory, order medicines, generate reports.
- **Administrator:** Full system control – user management, reporting, backup & restore, audit logs, reminders.

3.4 Reporting Requirements

- **Appointments Report:** Daily, weekly, and monthly appointment statistics.
- **Inventory Report:** Stock status, reorder list, low stock alerts.
- **Patient Report:** Prescription history, visit history.
- **System Report:** Audit logs, login activity, data backup logs.

4. Non-Functional Requirements

4.1.1 Performance and Load Requirements

For patient record management, inventory tracking, and appointment scheduling to go effectively, the Community Clinic Appointment and Inventory System must function well [1].

Current User Load: At first, it was built to support between 50 to 60 active users daily, including patients, doctors, administrative staff, and receptionists.

Expected Growth: As the hospital offers more services and more patients use the system, a 20–30% yearly increase in users is expected.

Number of Concurrent people: At least 20 people should be able to access the system at once without encountering any performance issues.

Transaction Size: Reports may include large data queries, although appointment scheduling, cancellations, and inventory updates are small .

Acceptable Maximum Average Transaction Time:

Appointment booking/cancellation takes ≤ 3 seconds, inventory updates ≤ 2 seconds, and report generation < 10 seconds, dependent on data amount.

UI Performance: On most devices, screen displaying and navigation responses should happen in less than two seconds

Scalability: Database and server scalability for increased loads in following phases should be supported by the system design.

4.1.2 Compatibility Requirements

The system needs to work with the clinic's technical structure and be accessible to patients using different devices. [2]

Supported HTML Versions: HTML5

Browser Versions Supported: Latest two stable releases of Chrome, Firefox and Edge

Database Versions Supported: MySQL as specified in the technology stack.

Communication Protocol: HTTPS for secure client-server communication

Frontend compatibility: Vue.js , Tailwind CSS

Supported Platform Versions: Cross-platform online application that functions on mobile browsers, Linux, and Windows.

External Frameworks or Standards:

For version control, use GitHub.

Manual testing or Selenium are used for validation.

Adherence to Sri Lanka's basic healthcare data confidentiality regulations.

4.1.3 External Interface Requirements

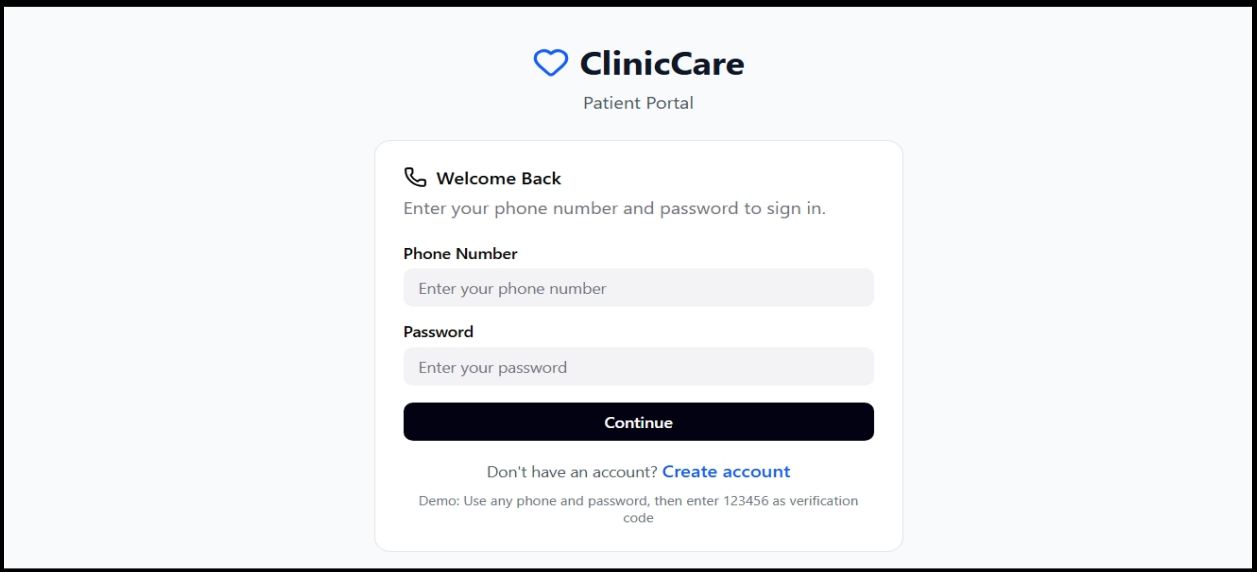
The Community Clinic Appointment and Inventory System requires multiple interfaces to ensure smooth communication between users, the database, and any connected external systems.

4.1.3.1. User Interface

The application will provide a **web-based, responsive user interface** accessible from desktops, laptops, tablets, and smartphones.

Patients

1. Patient's registration / log in page



The screenshot displays the 'ClinicCare Patient Portal' login interface. At the top, the 'ClinicCare' logo (a heart icon) and 'Patient Portal' text are centered. Below this, a white login box contains the following elements: a 'Welcome Back' heading with a bell icon, a prompt to 'Enter your phone number and password to sign in.', a 'Phone Number' label above a text input field with the placeholder 'Enter your phone number', a 'Password' label above a text input field with the placeholder 'Enter your password', a dark blue 'Continue' button, a link for 'Don't have an account? Create account', and a demo instruction: 'Demo: Use any phone and password, then enter 123456 as verification code'.




ClinicCare

Patient Portal

Create Account

Register to access our patient portal and manage your healthcare.

Full Name

Enter your full name

Email

Enter your email

Phone Number

Enter your phone number

Date of Birth

mm/dd/yyyy

Gender

Select gender

Password

Create a password (min. 6 characters)

Create Account

Already have an account? [Sign in](#)

2. Dashboard page – Overview of navigation




ClinicCare

Patient Portal

Navigation

Dashboard

Book Appointment

My Appointments

Medical Records

Profile

JD

John Doe

Patient

Logout

Dashboard

Welcome back, John Doe. Here's your health overview.

Quick Actions



Book Appointment

Schedule a new appointment

Book Appointment



View Appointments

Manage your appointments

View Appointments



Medical Records

Access your health records

Medical Records



Update Profile

Edit your personal information

Update Profile

Upcoming Appointments

Dr. Sarah Johnson

Cardiology

12/15/2024

10:00 AM

Consultation

View Details

Dr. Michael Chen

General Medicine

12/20/2024

2:30 PM

Follow-up

View Details

Notifications

Recent Updates



Lab results are ready for review

2 hours ago



Appointment reminder: Tomorrow at 10:00 AM

1 day ago

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SRS for Community Clinic Appointment and Inventory System

16

13.09.2023

3. Book Appointment page – Book appointment healthcare professionals

ClinicCare
Patient Portal

Navigation

- Dashboard
- Book Appointment**
- My Appointments
- Medical Records
- Profile

Book Appointment
Schedule your appointment with our healthcare professionals.

1. Select Doctor
Choose your preferred healthcare provider

- Dr. Sarah Johnson**
Cardiology
★ 4.9 15 years exp.
- Dr. Michael Chen**
General Medicine
★ 4.8 12 years exp.
- Dr. Emily Rodriguez**
Dermatology
★ 4.9 10 years exp.
- Dr. David Park**
Orthopedics
★ 4.7 18 years exp.

2. Select Date & Time
Choose your preferred appointment slot

Select Date:

September 2025

Su	Mo	Tu	We	Th	Fr	Sa
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4

Available Times:

9:00 AM	9:30 AM
10:00 AM	10:30 AM
11:00 AM	11:30 AM
2:00 PM	2:30 PM
3:00 PM	3:30 PM

3. Appointment Details
Specify the type of appointment

Appointment Type:
Select appointment type

Book Appointment

John Doe
Patient

Logout

4. Patient's appointment records page – View and manage patient appointment records

ClinicCare
Patient Portal

Navigation

- Dashboard
- Book Appointment
- My Appointments**
- Medical Records
- Profile

My Appointments
Manage your upcoming and past appointments.

Book New Appointment

Upcoming (2) Past (2)

Dr. Sarah Johnson **Scheduled**
Cardiology

Sunday, December 15, 2024
10:00 AM
Room 305, Main Building
Consultation

Notes: Regular checkup and heart monitoring

Reschedule **Cancel**

Dr. Michael Chen **Scheduled**
General Medicine

Friday, December 20, 2024
2:30 PM
Room 102, East Wing
Follow-up

Reschedule **Cancel**

Need Help?

If you need to reschedule or have questions about your appointment, please contact us:
Phone: +1 (555) 123-4567 **Email:** appointments@cliniccare.com **Hours:** Mon-Fri 8AM-6PM

John Doe
Patient

Logout

5. Patient's profile page – Personal information of pateint

The screenshot shows the 'Profile' page of the ClinicCare Patient Portal. The left sidebar contains a navigation menu with options: Dashboard, Book Appointment, My Appointments, Medical Records, and Profile (which is highlighted). Below the menu is a user profile for John Doe (JD) with a Logout button. The main content area is titled 'Profile' and includes an 'Edit Profile' button. It contains three sections: 'Personal Information' (Full Name: John Doe, Email: john.doe@email.com, Phone Number: 12345678, Date of Birth: 5/15/1990, Gender: Male, Address: 123 Main St, City, State 12345), 'Emergency Contact' (Emergency Contact Name: Not provided, Emergency Contact Phone: Not provided), and 'Medical Information' (Known Allergies: None reported).

ClinicCare
Patient Portal

Navigation

- Dashboard
- Book Appointment
- My Appointments
- Medical Records
- Profile**

JD John Doe
Patient

Logout

Profile

Manage your personal information and preferences.

Personal Information
Your basic personal details and contact information.

Full Name
John Doe

Email
john.doe@email.com

Phone Number
12345678

Date of Birth
5/15/1990

Gender
Male

Address
123 Main St, City, State 12345

Emergency Contact
Contact information for emergencies.

Emergency Contact Name
Not provided

Emergency Contact Phone
Not provided

Medical Information
Important medical history and current medications.

Known Allergies
None reported

Receptionists

6. Patient information page – Search patients details

The screenshot shows the 'Search Patients' page in the Reception Desk. The left sidebar contains a navigation menu with options: Dashboard, Search Patients (which is highlighted), and Book Appointment. Below the menu is a Logout button. The main content area is titled 'Search Patients' and includes a search bar with the placeholder text 'Search by name, phone, or email...'. Below the search bar is a list of 'All Patients (3)'. Each patient entry includes their name, phone number, email address, date of birth, and buttons for 'View Record' and 'Check In'.

Reception Desk

Dashboard

Search Patients

+ Book Appointment

Logout

Search Patients

Find patients by name, phone, or email

Search by name, phone, or email...

All Patients (3)

Sarah Johnson
(555) 123-4567 sarah.johnson@email.com DOB: 3/15/1985 View Record Check In

Michael Chen
(555) 234-5678 michael.chen@email.com DOB: 7/22/1978 View Record Check In

Emily Rodriguez
(555) 345-6789 emily.rodriguez@email.com DOB: 11/8/1992 View Record Check In

7. Appointment schedules page - Manage appointment schedules

The screenshot shows the 'Reception Desk' interface with a sidebar containing 'Dashboard', 'Search Patients', and 'Book Appointment'. The main area is titled 'Today's Schedule' for 'September 7, 2025'. It features three summary cards: 'Total Appointments' (3), 'Checked In' (1), and 'Completed' (0). Below these is a list of 'Today's Appointments' with three entries: Sarah Johnson (Cleaning) at 09:00 AM, Michael Chen (Consultation) at 10:30 AM, and Emily Rodriguez (Examination) at 02:00 PM. Each entry has buttons for 'Scheduled', 'Check In', and 'Cancel'. The 'Check In' button is active for Michael Chen. A 'Logout' button is in the sidebar, and a help icon is in the bottom right.

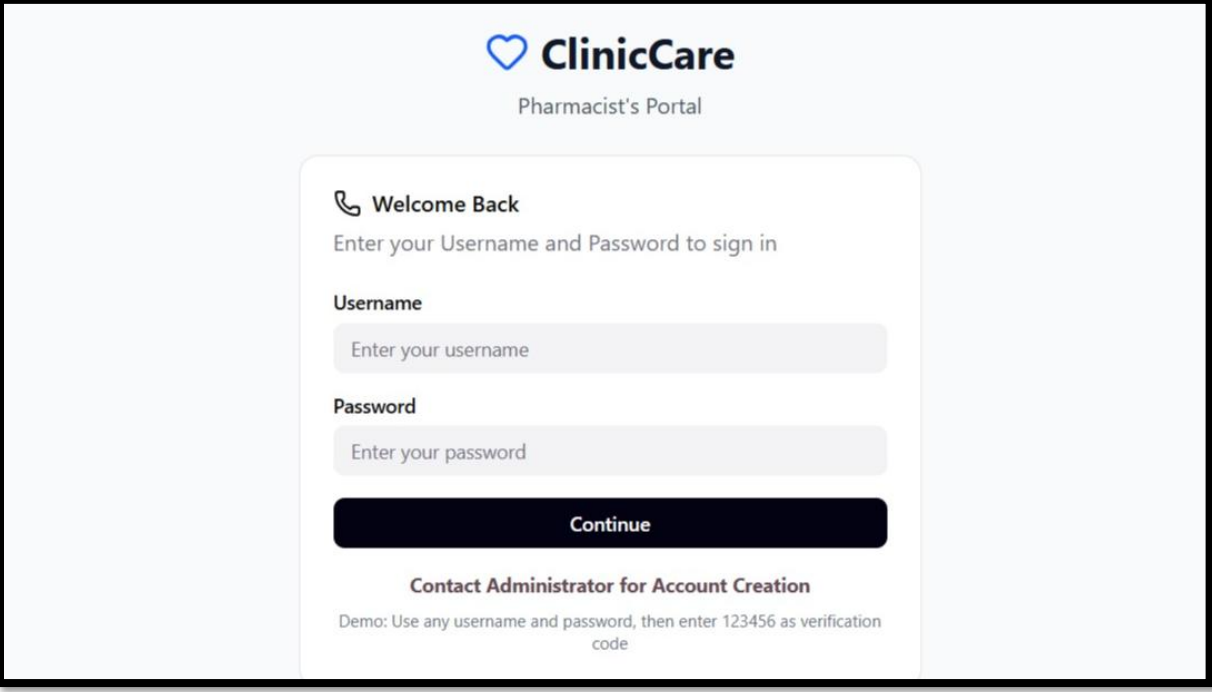
Time	Patient Name	Appointment Type	Status	Actions
09:00 AM	Sarah Johnson	Cleaning	Scheduled	Check In, Cancel
10:30 AM	Michael Chen	Consultation	Checked-In	Complete, Cancel
02:00 PM	Emily Rodriguez	Examination	Scheduled	Check In, Cancel

8. Book new appointment page – Schedule an appointment for patient

The screenshot shows the 'Book New Appointment' page in the 'Reception Desk' interface. The sidebar has 'Dashboard', 'Search Patients', and 'Book Appointment' (highlighted). The main area is titled 'Book New Appointment' with the subtitle 'Schedule an appointment for a patient'. It contains a form with the following fields: 'Appointment Details' (calendar icon), 'Patient *' (dropdown with 'Select a patient'), 'Date *' (calendar icon, format 'mm/dd/yyyy'), 'Time *' (clock icon, dropdown with 'Select time'), 'Appointment Type *' (dropdown with 'Select appointment type'), and 'Notes' (text area with placeholder 'Additional notes for the appointment...'). At the bottom are 'Book Appointment' and 'Cancel' buttons. A help icon is in the bottom right.

Pharmacist

9. Pharmacist's log in page



The screenshot shows the login interface for the ClinicCare Pharmacist's Portal. At the top, the ClinicCare logo (a blue heart icon) and the text "ClinicCare" are displayed, followed by "Pharmacist's Portal" in a smaller font. Below this, a white rounded rectangle contains the login form. The form starts with a "Welcome Back" message and a prompt to "Enter your Username and Password to sign in". It features two input fields: "Username" with the placeholder "Enter your username" and "Password" with the placeholder "Enter your password". A dark blue "Continue" button is positioned below the password field. At the bottom of the form, there is a link "Contact Administrator for Account Creation" and a demo instruction: "Demo: Use any username and password, then enter 123456 as verification code".

ClinicCare
Pharmacist's Portal

Welcome Back
Enter your Username and Password to sign in

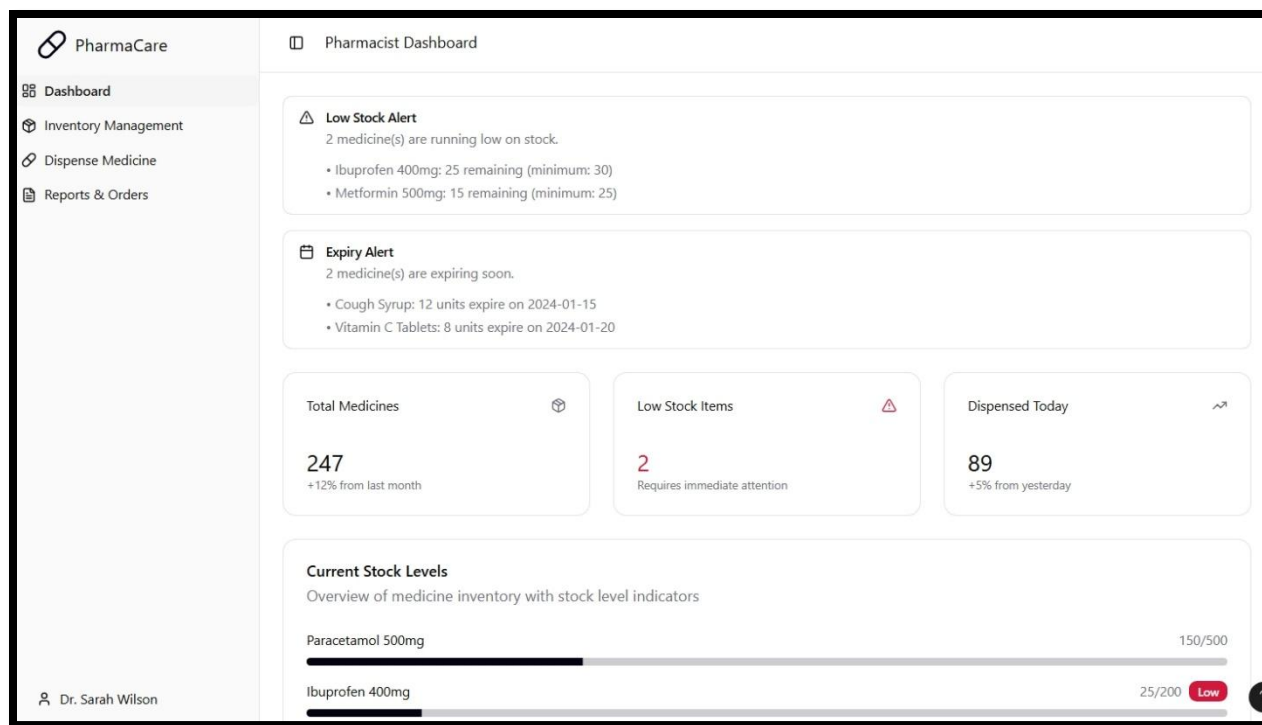
Username
Enter your username

Password
Enter your password

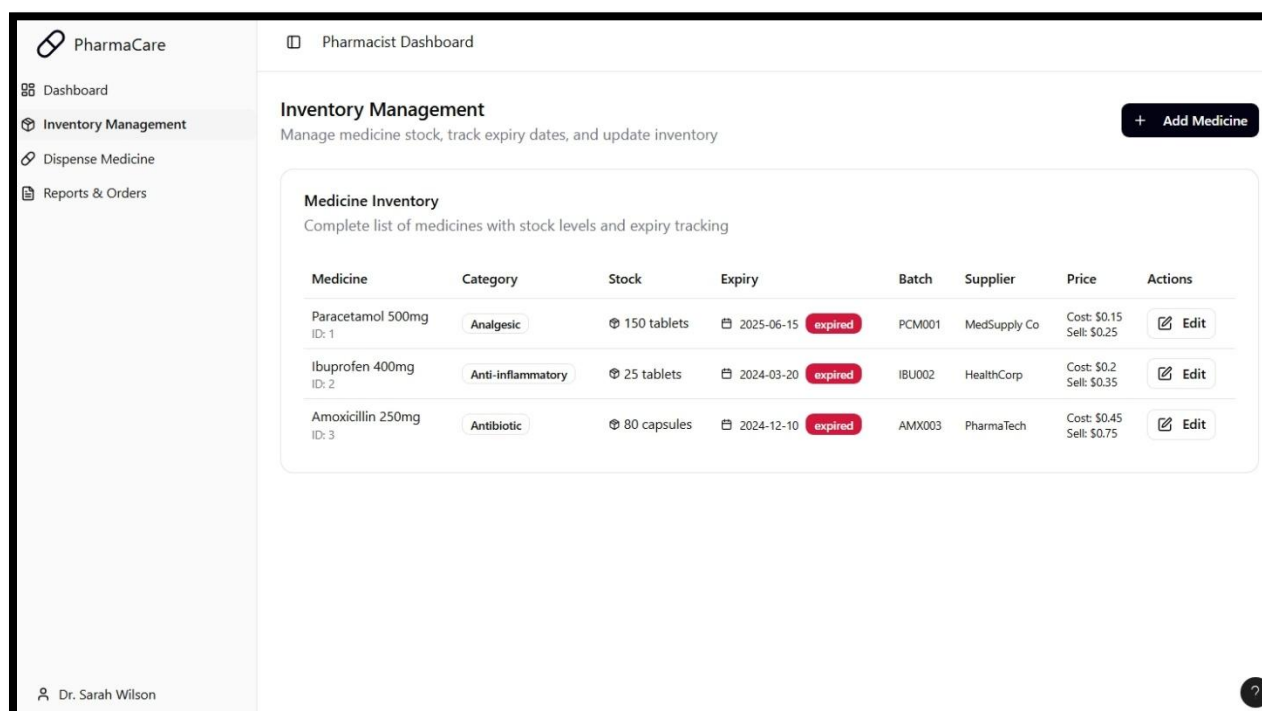
Continue

Contact Administrator for Account Creation
Demo: Use any username and password, then enter 123456 as verification code

10. Pharmacist dashboard page



11. Inventory management page – Manage medicine stock, track expiry dates and update inventory



12. Medicine dispense page – View doctor prescriptions and dispense medicine to patients

- Dashboard
- Inventory Management
- Dispense Medicine
- Reports & Orders

Pharmacist Dashboard

Dispense Medicine

View doctor prescriptions and dispense medicines to patients

Search Prescriptions

Search by patient name, prescription ID, or doctor

Filter by Status

All Status

Doctor Prescriptions

Active prescriptions requiring medicine dispensing

Prescription ID	Patient	Doctor	Date	Medicines	Status	Actions
RX001	John Smith ID: P12345	Dr. Emily Johnson	2024-01-08	Paracetamol 500mg (30) Ibuprofen 400mg (20)	Pending	Dispense
RX002	Sarah Wilson ID: P67890	Dr. Michael Brown	2024-01-08	Amoxicillin 250mg (21)	Pending	Dispense
RX003	Robert Davis ID: P11223	Dr. Lisa Anderson	2024-01-07	Aspirin 75mg (90)	Dispensed	Completed

Dr. Sarah Wilson

13. Reports and orders page – Generate reports and manage medicine orders

- Dashboard
- Inventory Management
- Dispense Medicine
- Reports & Orders

Pharmacist Dashboard

Reports & Orders

Generate reports and manage medicine orders

Reports

Manage Orders

Generate Reports

Create detailed reports for stock analysis and management

Report Type

Stock Usage Report

From Date

01/01/2024

To Date

01/08/2024

Generate Report

Stock Usage Report Preview

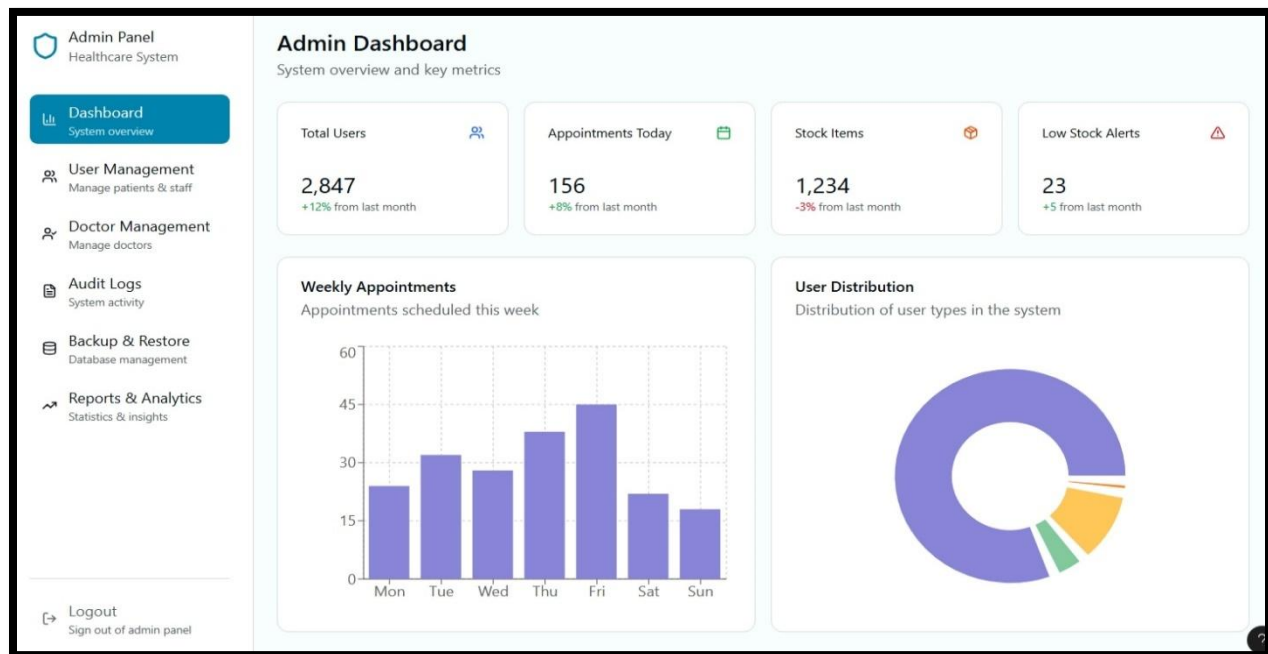
Current week stock movement and reorder recommendations

Medicine	Dispensed	Remaining	Reorder Level	Status	Action
Paracetamol 500mg	45	150	50	Good	
Ibuprofen 400mg	35	25	30	Low Stock	Order
Amoxicillin 250mg	28	80	40	Good	
Aspirin 75mg	15	200	60	Good	
Metformin 500mg	42	15	25	Low Stock	Order

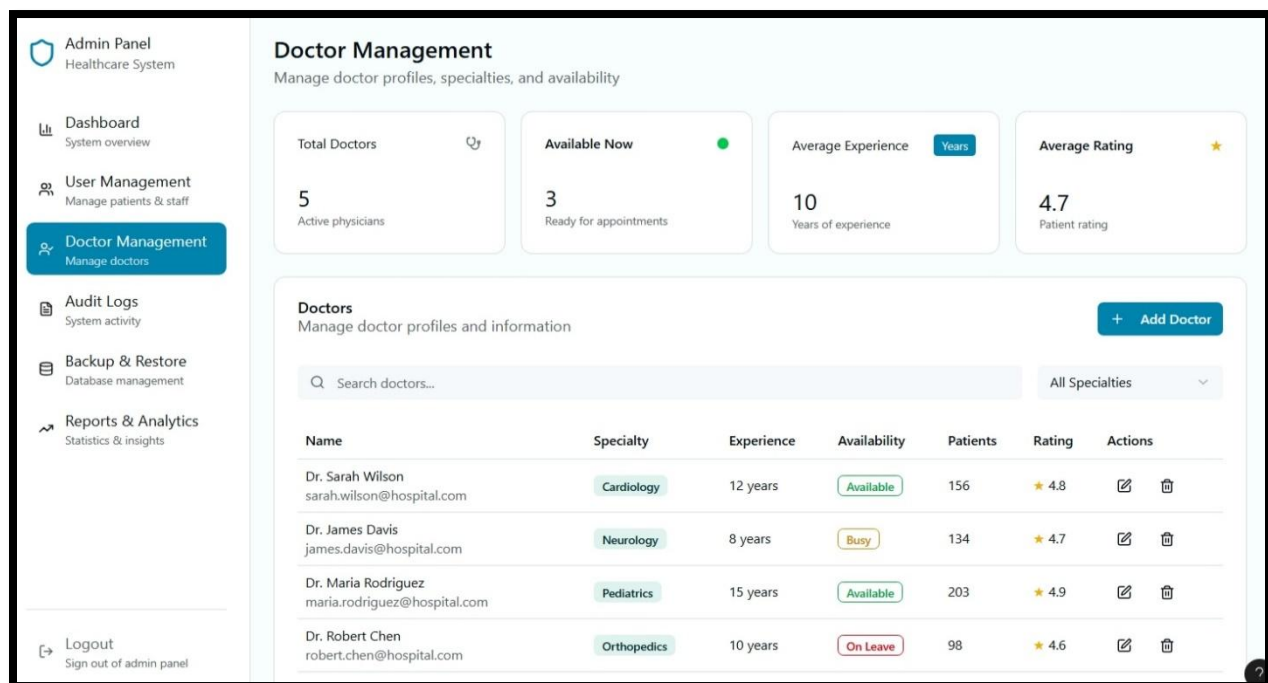
Dr. Sarah Wilson

Admin - Manage users, inventory, and generate reports.

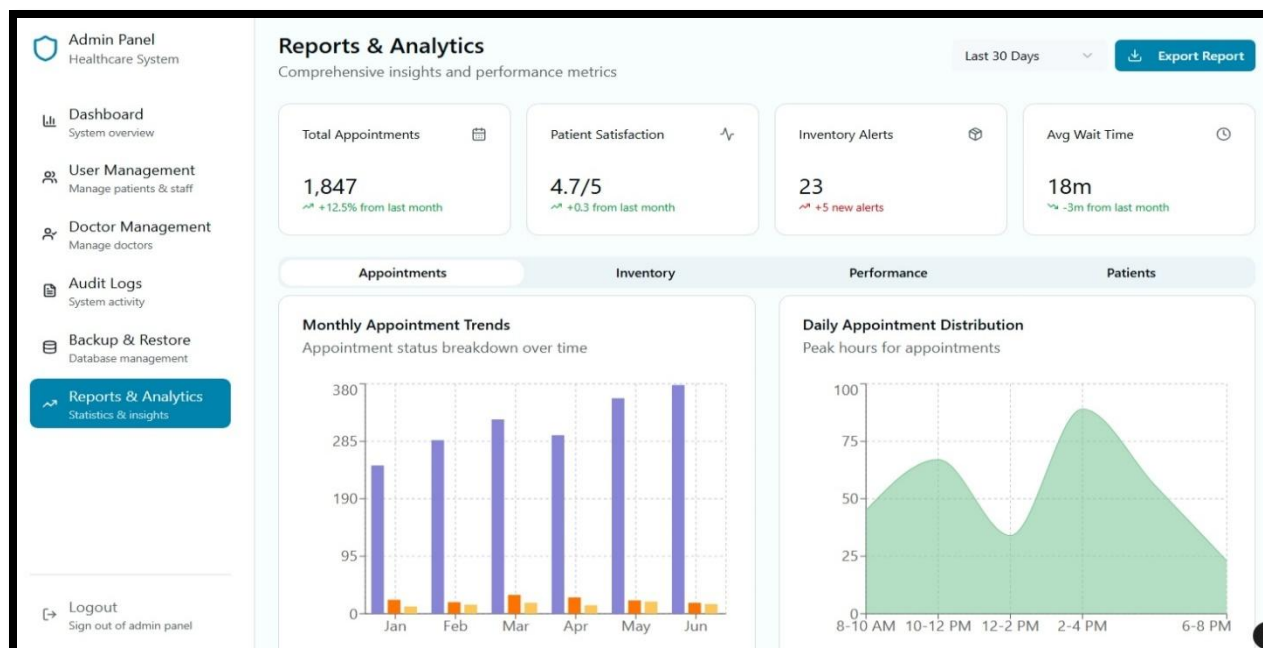
14. Admin dashboard page – System overview and analytics



15. Doctor details and schedule page – Manage doctor profile and availability



16. Reports and analytics page - Generate reports and view analytics



17. User management page – Manage user accounts

Admin Panel
Healthcare System

User Management
Manage patients, doctors, and staff members

Users
View and manage all system users [+ Add User](#)

Search users... All Types

Name	Email	Type	Status	Last Login	Actions
John Smith	john.smith@example.com	Patient	Active	2024-01-15	View Edit Delete
Dr. Sarah Wilson	sarah.wilson@hospital.com	Doctor	Active	2024-01-16	View Edit Delete
Emily Johnson	emily.johnson@hospital.com	Staff	Active	2024-01-14	View Edit Delete
Michael Brown	michael.brown@example.com	Patient	Inactive	2024-01-10	View Edit Delete
Dr. James Davis	james.davis@hospital.com	Doctor	Active	2024-01-16	View Edit Delete
Lisa Garcia	lisa.garcia@hospital.com	Staff	Active	2024-01-15	View Edit Delete
Robert Miller	robert.miller@example.com	Patient	Active	2024-01-13	View Edit Delete
Amanda Wilson	amanda.wilson@example.com	Patient	Active	2024-01-16	View Edit Delete

Logout
Sign out of admin panel

Doctor - Access to patient schedules and past records

Rural Siddha Hospital Thellipalai
Copyright © 2004 Quality Software (Pvt) Ltd.

SRS for Community Clinic Appointment and Inventory System

18. Doctor login page

**ClinicCare**
Doctors Portal

 **Welcome Back**

Enter your Username and Password to sign in

Username

Password

Continue

Contact Administrator for Account Creation
Demo: Use any username and password, then enter 123456 as verification code

19. Prescription management page – Manage digital prescriptions

MedDoc Portal

DashboardConsultationPrescriptions

DSJ Dr. Sarah Johnson
Internal Medicine

Prescription Management
Create and manage digital prescriptions

Manage Prescriptions

Create New

All Status

John Smith Sent

Medication
Lisinopril

Dosage
10mg

Frequency
Once daily

Duration
30 days

Instructions
Take with food in the morning

Date: 2024-09-08 • Pharmacy: CVS Pharmacy

View Details

Maria Garcia Filled

Medication
Metformin

Dosage
500mg

Frequency
Twice daily

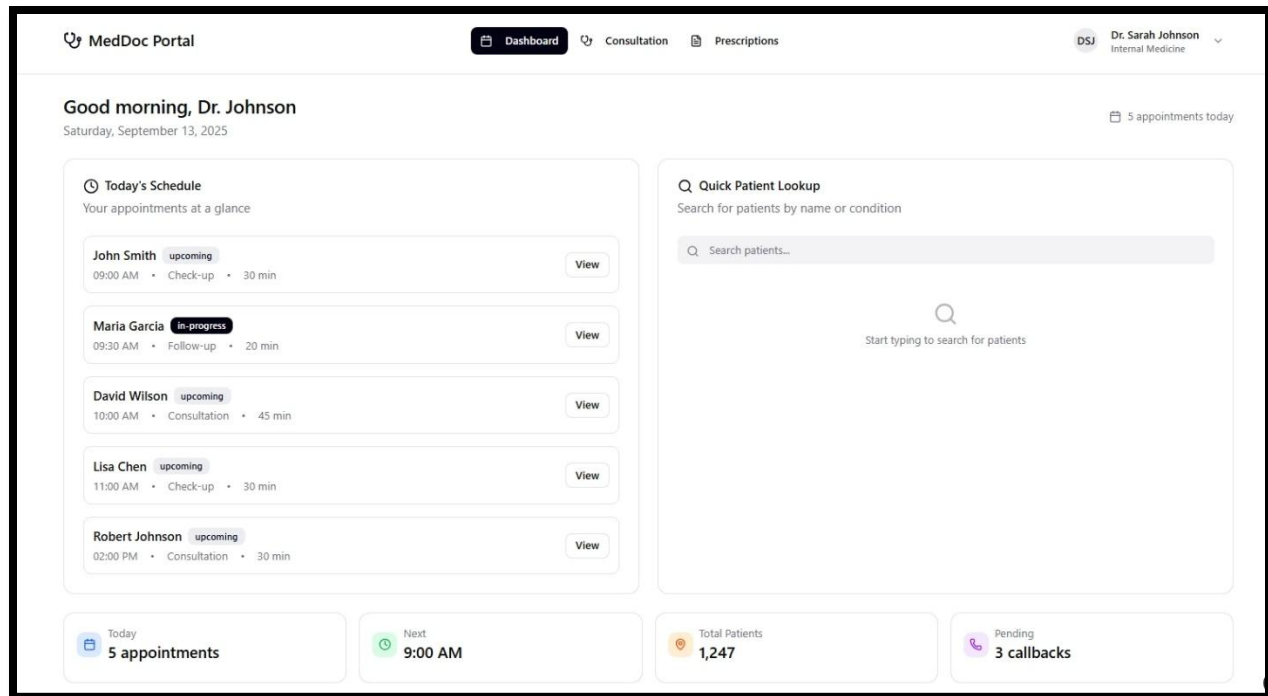
Duration
90 days

Instructions
Take with meals

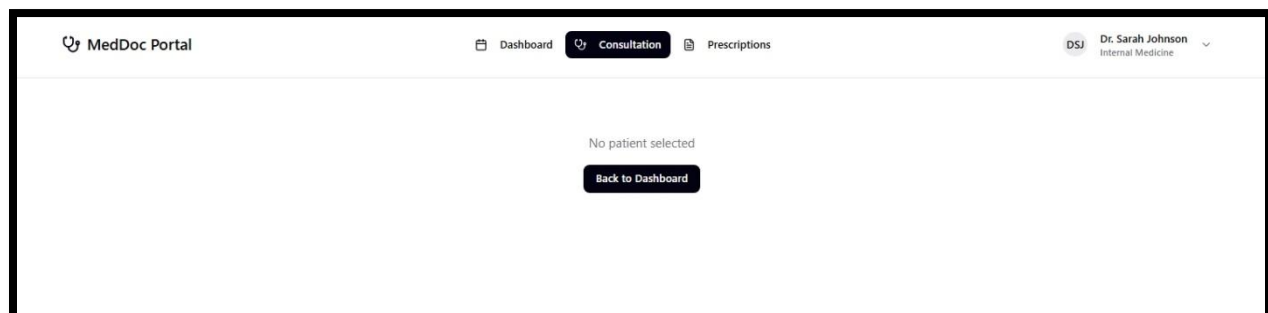
Date: 2024-09-07 • Pharmacy: Walgreens

View Details

20. Doctor profile page – View patient schedules



21. Consultation page – View selected patients



Special Requirements:

- Support for bilingual interface (English and Tamil/Sinhala) for accessibility.
- Minimalist UI for rural users with low digital literacy.
- Responsive layout ensuring screen load time ≤ 2 seconds on standard devices.

4.1.3.2. Hardware Interface [4]

Server specifications:

The application requires a web server environment with MySQL and Node.js, and minimum hardware specifications of a quad-core CPU, 8 GB of RAM, and 100 GB of storage.

Client Devices:

Any modern web-connected device, such as tablets, smartphones, desktops, and laptops, is supported, provided it meets a minimum resolution of 1024×768 .

The edges:

The proposed system includes printers for generating prescriptions and reports, with future integration planned for medicine stock barcode scanners.

4.1.3.3. Software Interface [3]

Frontend: Vue.js with Tailwind CSS for responsive UI.

Backend: Node.js with Express.js to manage APIs and server logic.

Database: MySQL is utilized for structured patient data, appointment scheduling, and inventory management, while MySQL Workbench serves for query management and layout design.

Version Control: A collaborative development resource on GitHub.

Testing tool: Selenium is a testing tool for automated user interface testing.

External APIs: The system includes external APIs for email and SMS notifications to remind patients of appointments, as well as a potential payment gateway interface for charged consultations.

4.1.3.4. Communication Interface

For secure client-server communication, HTTPS protocols are utilized. Internal communication between the Vue.js frontend and Node.js/Express backend is facilitated by HTTP/REST APIs. [3]

The system utilizes the clinic's local area network and is accessible via the public internet.

Security measures include TLS 1.3 encryption for communications and role-based login for user authentication.

4.1.4 Security and Authentication requirements

4.1.4.1 Data Storage Security

Using access control based on roles, MySQL is used to store patient, appointment, and inventory data. The algorithm AES-256 is used to encrypt sensitive data while it is at rest, including patient information and medical history. [6]

Assigning users to roles (Patient, Doctor, Receptionist, Administrator) and providing permissions according to the least priority principle manage access control.

A backup and recovery plan is described in the text that is provided. This method includes a recovery plan aimed at restoring data in the case of damage or system failure, as well as daily automated database backups to safe storage.

4.1.4.2 Data Communication Security

HTTPS over TLS 1.3 must be used to encrypt communication between the frontend and backend.

Sensitive information is not saved in browser caches or URLs; instead, POST requests are used for sensitive functions like login and appointment scheduling.

After a certain amount of time of absence, user sessions are automatically logged out and managed by encrypted tokens, specifically JWT. [7]

4.1.5 Quality Assurance Requirements

4.1.5.1 QA Test Scope

Unit Testing: Verification of individual modules (e.g., appointment booking, inventory update).

Integration Testing: Ensuring communication between modules (frontend, backend, and database) works as intended.

System Testing: Full end-to-end workflow testing (appointment scheduling, stock alerts, reporting).

User Acceptance Testing (UAT): Testing by clinic staff and selected patients to validate usability and requirements.

Performance & Load Testing: Evaluating system response under concurrent user loads.

Security Testing: Checking for vulnerabilities such as unauthorized access, SQL injection, and data leakage. [8]

4.1.5.2 QA Environment

Three software tools are listed in the provided document: MySQL Workbench for database query validation, Postman for API testing, and Selenium for automated user interface testing.

A server with a quad-core CPU, 8 GB of RAM, and 100 GB of storage makes up the hardware environment. Smartphones, tablets, laptops, and desktop computers are examples of client devices.

In according to Continuous Integration/Continuous Development principles, the workflow defined makes use of GitHub Actions to automate build and test cycles and GitHub for version control.

4.1.5.3 QA Data

Patient records with descriptive and medical history, appointment scheduling and cancellation information, and inventory data about medications and supplies with inventories and expiration dates are among the data entities that are supplied.

The given content makes reference to other data sources, including false notification information more especially, email and SMS reminders.

Repeatable testing scenarios are made possible by scripts that may automatically generate test appointments, patient records, and inventory entries.

4.1.6 Development Requirements

4.1.6.1 Development Environment

The software platform includes JavaScript for the frontend, HTML5, Tailwind CSS, Vue.js, and Node.js with Express.js for the backend REST API. GitHub is used for version control, while MySQL Workbench is used to administer the database. Postman is used for API testing, whereas Selenium is used for user interface testing. [9]

The minimum specifications for developer workstations are 500 GB of storage, 8 GB of RAM, and a quad-core CPU. A Linux or Windows server with the Node.js runtime installed is required for testing as a local development server.

Integrated Development Environments (IDEs) such as Visual Studio Code , along with the npm package manager for installing and managing dependencies, are part of the suggested environment setup.

4.1.6.2 Development Data

Key Data Entities for Development:

- Patients: Dummy patient records with demographics and medical history.
- Appointments: Sample booking/rescheduling/cancellation data.
- Doctors/Staff: Dummy schedules and user accounts.
- Inventory: Sample medicines, stock levels, and expiry dates.

External Data Sources:

- Mock SMS/email data for notification testing.

Data Generation Tools:

- SQL scripts for creating and populating sample databases.
- Automated JSON/CSV test datasets for bulk upload testing.

4.1.6.3 Coding Standards

Naming Conventions:

- Variables & Functions: `bookAppointment()`, `inventoryLeve()`
- Classes & Components: `AppointmentController`, `InventoryManager`
- Database Tables: `patient_records`, `appointment_logs`

Coding Guidelines:

- Follow JavaScript ES6+ standards.
- Use linting tools to enforce consistent style.
- Modular structure for reusability and scalability of components.
- Inline documentation using JSDoc for functions and API endpoints.

Best Practices:

- Separation of concerns between frontend and backend.
- Secure coding practices

4.1.6.4 Implementation Packaging Requirements

The application is made up of a Node.js backend that acts as an Express.js REST API, a Vue.js frontend that is packaged into static files using Vite/Webpack, and SQL scripts for database schema management and migrations. [10]

GitHub tags are used to handle versioning, which organises the deployment structure (e.g., v1.0.0). `/frontend` (Vue.js), `/backend` (Express.js), `/database` (SQL), and `/tests` are important folders. An upcoming portability enhancement is Docker containerisation.

Special Requirements: To facilitate simple maintenance, modules must be loosely connected and autonomous. OpenAPI/Swagger-documented APIs for external integration.

4.1.7 Deployment Requirements

4.1.7.1 Installation Packaging Requirements

The frontend of the application is built up of packaged Vue.js static assets (HTML, CSS, and JS) that were constructed with Vite/Webpack. A Node.js Express server that is controlled by npm scripts serves as the backend. MySQL schema and seed data supplied as SQL scripts are examples of database components.

Deployment scripts for Linux-based servers are available, and future installations will make use of Docker containers for a more simple environment setup.

4.1.7.2 Deployment Requirements

The hospital's local server is where the application is installed, and cloud backup is an option. A quad-core CPU, 8 GB of RAM, and 100 GB of storage are the minimum requirements for a system. [9]

Patients use secure HTTPS to access the system using desktop and mobile browsers, while doctors, administrative staff, and receptionists use both local area network (LAN) and remote internet connections.

For servers, the system supports Windows Server 2019 or Linux Ubuntu 20.04 LTS. Windows 10/11, Android 10 and later, iOS 14 and later, and macOS 11 and later are all suitable with the client.

At any given time, only one version of the software is being actively used. GitHub keeps older versions to make backups easier if needed.

4.1.7.3 Documentation requirements

Standard software development documentation, such as the Software Requirements Specification (SRS), System Design Document (SDD), Test Plans and Test Reports, and Deployment and Maintenance Guides, are provided in the list.

In addition to instant start guidelines for scheduling and cancelling appointments and updating inventory, end-user documentation includes a comprehensive user manual for patients, doctors, and administrators.

This technical documentation includes a comprehensive database schema with Entity-Relationship (ER) diagrams and API specifications created using Swagger standards.

4.1.8 Special Documentation Requirements

This system serves as a tool for hospital information, specifically missing emergency or diagnostic services.

The document provided describes a project's copyright and licencing conditions. The MIT Licence governs internal use of the source code. Furthermore, the open-source licences for Vue.js, Tailwind CSS, and Node.js, in particular, regulate how these third-party libraries are used. [9]

The interface design needs to include hospital branding, such as the logo and colour scheme.

4.1.9 Applicable Standards

Compliance with Sri Lankan Ministry of Health guidelines for patient data confidentiality is a key legal and regulatory standard. [13]

The communication standards identified are TCP/IP for networking and HTTPS with TLS 1.3 for secure communication.

The platform adheres to cross-browser compatibility standards, supporting the latest versions of Chrome, Firefox, Edge, and Safari. It also provides operating system support for Windows, Linux, Android, and iOS.

The document outlines two quality and safety standards: ISO/IEC 25010:2011, a software quality model covering usability, reliability, and maintainability, and ISO/IEC 27001:2018, detailing information security management best practices.

4.1.10 On-line User Documentation and Help System Requirements

The system is designed to offer comprehensive support through an Online Help Section featuring Frequently Asked Questions (FAQs) and step-by-step guides. It will also incorporate "Help/About" notices directly within the web application. Context-sensitive tooltips will be available for input fields, such as those found on appointment booking forms, to aid users. Furthermore, future integration of chatbot based support for addressing patient queries is a possibility. [11]

4.1.11 Usability Requirements

Browser Compatibility: Must work on the latest two stable versions of Chrome, Firefox, Edge, and Safari.

Accessibility:

Support for WCAG 2.1 accessibility guidelines (e.g., screen readers, high-contrast mode).

Multilingual interface (English, Tamil, Sinhala).

UI/UX Design:

Minimalist design with clear navigation for rural users with low digital literacy.

Mobile-responsive layout for smartphones.

Usability Testing:

Conducted during UAT (User Acceptance Testing) with actual patients and staff.

Feedback loop included to refine UI elements. [11]

Appendix

A. Acronyms and Abbreviations

- **SRS** – Software Requirements Specification
- **UI** – User Interface
- **RBAC** – Role-Based Access Control
- **HTTPS** – Hypertext Transfer Protocol Secure
- **TLS** – Transport Layer Security
- **API** – Application Programming Interface
- **ER Diagram** – Entity Relationship Diagram

B. Sample Data Structures

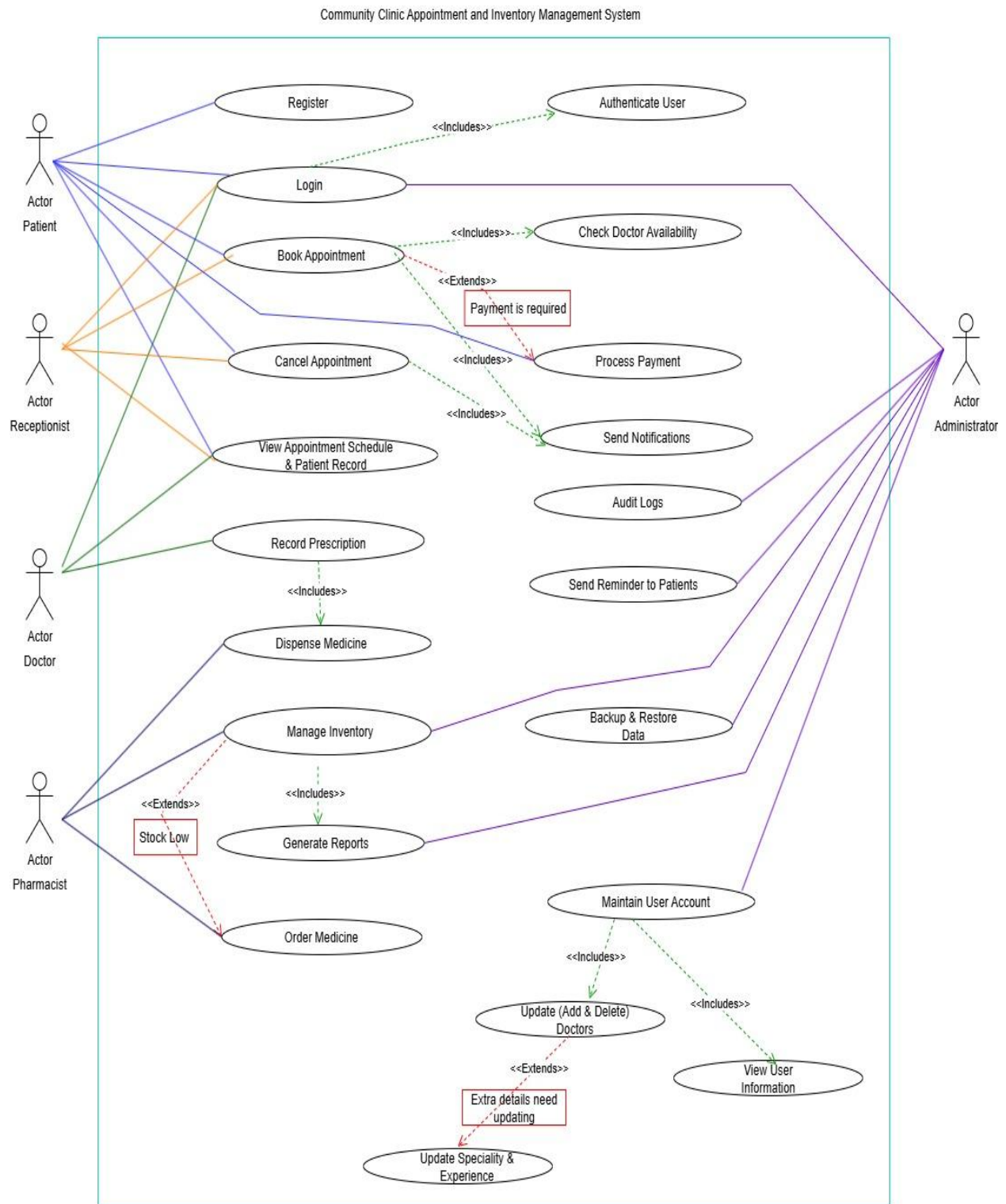
1. Patient Table (Database Schema Example)

Field Name	Data Type	Description
patient_id	INT (PK)	Unique identifier for patient
name	VARCHAR(100)	Full name
age	INT	Patient's age
gender	VARCHAR(10)	Male / Female / Other
contact_no	VARCHAR(15)	Phone number (unique)
address	VARCHAR(255)	Residential address
email	VARCHAR(100)	Patient email (unique)

2. Appointment Table

Field Name	Data Type	Description
appointment_id	INT (PK)	Unique appointment number
patient_id	INT (FK)	Links to Patient table
doctor_id	INT (FK)	Assigned doctor
appointment_date	DATE	Scheduled date
status	VARCHAR(20)	Booked / Cancelled / Completed

C. Use Case Diagram



D. Testing Checklist

- **Functional Tests:** Appointment booking, cancellation, prescription recording, inventory update.
- **Non-Functional Tests:** Load testing (20 concurrent users), security testing (SQL injection, XSS), usability testing with staff.
- **User Acceptance Tests (UAT):** Conducted with real clinic staff and selected patients.

E. Project Timeline (Summary)

Phase	Task	Duration	Dates
Week 1–2	Requirement Gathering & Analysis	2 weeks	1st Aug – 14th Aug 2025
Week 3–4	System Design	2 weeks	15th Aug – 28th Aug 2025
Week 5–8	Backend & Frontend Development	4 weeks	29th Aug – 25th Sept 2025
Week 9–10	Integration & Testing	2 weeks	26th Sept – 9th Oct 2025
Week 11	User Training & Deployment	1 week	10th Oct – 16th Oct 2025
Week 12	Final Review & Documentation	1 week	17th Oct – 23rd Oct 2025

Approval Letter from Client

Head Doctor
Rural Siddha Hospital, Tellipalai
Tellipalai, Jaffna District, Sri Lanka
24 /08/2025

Undergraduates,
Bachelor of Software Engineering
Department of Electrical and Computer Engineering,
Open University of Sri Lanka.
29/08/2025

Subject: Confirmation of Collaboration for Web-Based Clinic Appointment and Inventory System

Dear Mr. Thilaharaj, 321435714,

I am writing on behalf of Rural Siddha Hospital, Tellipalai to formally acknowledge and express our appreciation for your initiative to develop a Web-Based Clinic Appointment and Inventory Management System for our institution.

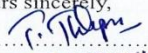
Our hospital has long faced challenges in managing patient appointments, doctor schedules, and medicine inventory efficiently. We believe that the proposed system will significantly improve our workflow, reduce delays, and enhance the quality of service provided to our patients.

We hereby confirm our willingness to act as the client organization for this project. We will provide access to relevant operational details, requirements, and feedback to ensure the successful design and implementation of the system.

We look forward to working closely with you and your team and will extend our full support throughout the development process.

Thank you once again for selecting our hospital for this project.

Yours sincerely,

Dr. 
Medical Officer in Charge
Rural Siddha Hospital
Tellipalai